Conduction Heat Transfer Tutorial

1 Questions

1. One face of a copper plate 10 cm $\times$ 25 cm $\times$ 3cm thick is maintained at 400 ℃. The other face is maintained at 100 ℃. Calculate the heat transfer rate through the plate (Ans. 92.5 kW).

2. An exterior of a house may be approximated by a 10.2 cm layer of common brick [k= 0.7 W/m ℃] followed by a 3.8 cm layer of gypsum [k= 0.48 W/m ℃]. What thickness of loosely packed rock wool insulation [k= 0.065 W/m ℃] should be added to reduce the heat loss (or gain) through the wall by 80 % ?[Q-with insulation / Q with out insulation=0.2]. (Ans. 0.0584 m).

3. A thick walled tube of stainless steel [k= 19 W/m ℃] with 2 cm inner diameter and 4cm outer diameter is covered with a 3cm layer of asbestos insulation [k= 0.2 W/m ℃]. The inside wall temperature of the pipe is maintained at 600 ℃, and outside surface temperature is 100 ℃. Calculate the heat loss per meter of length. Also calculate the tube insulation interface temperature. (Ans. 680 W/m. 595.8 ℃)

4. Calculate the critical radius of insulation for asbestos [k=0.17 W/mK] surrounding a pipe and exposed to room air at 20 ℃with h 3 W/m^2K. Calculate the heat loss from a 200 ℃, 5 cm diameter pipe when covered with the critical radius of insulation and without insulation. [Ans. Q_{criti}= 105.7 W/m, without 85 W/m]

5. A slab of thickness 4 cm is subjected to a hot air flow at 130 ℃with h 500 W/m^2K and cold air at 30 ℃with h 250 W/m^2K on each side. The plate thermal conductivity is 20 W/m ℃. Estimate heat transfer through the plat per meter square. [Ans 12.5 kW]

6. A current of 200 A is passed through a stainless steel wire [k=19 W/mK] 3 mm in diameter. The resistivity of steel may be assumed as 70 $\mu\Omega$ cm, and the length of the wire is 1 m. The wire is submerged in a liquid at 110 ℃and experience a convection heat transfer coefficient of 4 kW/m^2℃. Calculate the centre temperature of the wire. (Ans 231.6 ℃)

7. An aluminum rectangular fin[k= 200 W/m℃] 3 mm thick and 7.5 cm long protrudes from a wall. The base is maintained at 300 ℃. And the ambient temperature is 50 ℃, with h 10 W/m^2K. Calculate the heat loss from the fin per unit depth of material. [Ans. 360 W/m]

8. Aluminum fins 1.5 cm wide and 1 mm thick are placed on a 2.5 cm diameter tube to dissipate the heat. The tube surface is 170 ℃, and the ambient fluid temperature is 25 ℃. Calculate the heat loss per fin for h 130 W/m^2K. Assume k = 200 W/mK for aluminum. [ans. 61 W].

9. A steel ball [c = 0.46 kJ/kg ℃, k = 35 W/m · ℃] 5.0 cm in diameter and initially at a uniform temperature of 450 ℃is suddenly placed in a controlled environment in which the temperature is maintained at 100 ℃. The convection heat-transfer coefficient is 10 W/m 2 ·℃. Calculate the time required for the ball to attain a temperature of 150 ℃.

10. A wall 2 cm thick and thermal conductivity 1.3 W/mK is to be insulated with an average thermal conductivity 0.35 W/mK. Find the thickness of the insulation so that the heat loss per square meter will not exceed 1830 W. Assume the inner and outer surface temperatures are 1300 ℃, and 30℃, calculate the thickness of the insulation.

11. A certain material 2.5 cm thick, with a cross-sectional area of 0.1 m^2, has one side at 35 ℃, and the other at 95 ℃. The centre is at 62 ℃, and the heat flow through the material is 1 kW. Obtain an expression for the thermal conductivity of the material as a function of temperature.

12. One side of a copper block 5 cm thick is at 250 ℃, The other side is covered with a layer of fibre glass 2.5 cm thick. The outside of the fibre glass is maintained at 35 ℃, and the total heat flow through the slab is 52 kW. What is the area of the slab.

13. A plane wall is with a material having thermal conductivity that varies as the square of the temperature according to the relation $k = k_0(1 + \beta T^2)$. Derive an expression for the heat transfer in such a wall.

14. A material has a thickness of 30 cm and a thermal conductivity of 0.04 W/mK. At a particular instant in time the temperature distribution with x, the distance from the left face, is $T = 150x^2 - 30x$, where x is in meters. Calculate the heat flow rate at x = 0, and x = 30 cm. Is the solid heating up or cooling down ?

15. In a quenching process steel rods ($\rho = 7832 kg/m^3$, $c_p = 434 J/kgK$, k= 63.9 W/mK) are heated in a furnace to 850 ℃and then cooled in a water bath to an average temperature of 95 ℃. The water bath has a uniform temperature of 40 ℃and convection heat transfer coefficient of 450 W/m^2K. If the steel rods have a diameter of 50 mm and length of 2m, determine the time required to cool a steel rod from 850 ℃to 95

℃in the water bath and total amount of heat transfer during this. (Bi 0.088, t =254 s, Q= 10.1 MJ).

16. Estimate the minimum depth at which one must place a water main below the surface to avoid freezing. The soil is initially at a uniform temperature of 20 ℃. Assume that under the worst conditions anticipated it is subjected to a surface temperature of -15 ℃for a period of 60 days. Use the following properies for soil. ($\rho = 2050kg/m^3, k = 0.52W/mK, c_p = 1840J/kgK, \alpha = 0.138 \times 10^{-6}m^2/s$) [Ans 0.68 m]

17. A large block of steel [$k = 45/WmK, \alpha = 1.4 \times 10^{-5}m^2/s$] is initially at a uniform temperature of 35 ℃. The surface is exposed to a heat flux by suddenly raising the surface temperature to 250 ℃. Calculate the temperature at a depth of 2.5 cm after a time of 0.5 min. (Ans. 118.5 ℃)

18. A horizontal pipe 15cm diameter and 4 m long is buried in the earth at a depth of 20 cm. The pipe wall temperature is 75 ℃, and the earth surface temperature is 5 ℃. Assuming that the thermal conductivity of the earth is 0.8 W/mK, calculate the heat lost by the pipe. (Ans. 860 W)

19. A small cubical furnace 50 cm by 50 cm by 50cm on the inside is constructed of fireclay brick [k = 1.04 W/mK] with a wall thickness of 10 cm. The inside of the furnace is maintained at 500 ℃, and the outside is at 50 ℃. Calculate the heat lost through the walls. [Ans. 8.59 kW]